

Airbnb Marketplace Intelligence Platform

Business Questions & Project Roadmap

This document outlines the business questions that each stage of the end-to-end analytics project is designed to answer. The workflow mirrors a real-world analytics engagement, where raw data is transformed into actionable business insights using Excel, SQL, Python, and Power BI.

1. Excel – Data Audit & Initial Exploration

1. Is the raw data complete, accurate, and ready for analysis?
2. Which columns contain missing, inconsistent, or duplicate values?
3. How are listings distributed across cities, neighbourhoods, property types, and room types?
4. What are the initial pricing patterns across different markets?
5. Are there obvious anomalies or outliers in prices, ratings, or host information?
6. What high-level trends can be identified before deeper analysis?

2. SQL – Data Preparation & Business Analysis

1. Which cities and neighbourhoods have the largest Airbnb marketplaces?
2. Which markets generate the highest customer demand based on review activity?
3. How do property type, room type, and accommodation capacity influence pricing?
4. Which hosts consistently outperform others in ratings and responsiveness?
5. Do Superhosts deliver better guest experiences than regular hosts?
6. How does host experience influence listing performance?
7. Which listings are significantly overpriced or underpriced compared to local market averages?
8. What seasonal trends exist in customer reviews and booking activity?
9. Which neighbourhoods contribute the largest share of marketplace demand (Pareto analysis)?
10. What KPIs should executives monitor to evaluate marketplace health?

3. Python – Exploratory Data Analysis & Statistical Insights

1. What relationships exist between pricing, ratings, host characteristics, and customer demand?
2. Which variables have the strongest influence on listing prices?
3. Are differences between Superhosts and Non-Superhosts statistically significant?
4. Do Instant Book listings receive higher ratings or greater demand?
5. What hidden trends and outliers are not visible through SQL summaries alone?
6. What new analytical features can improve business understanding (e.g., Price per Guest, Host Experience, Review Density)?

4. Power BI – Executive Business Intelligence Dashboard

1. What is the overall health of the Airbnb marketplace across all cities?
2. Which markets present the greatest growth opportunities?
3. Which listings require operational attention?
4. How do pricing, quality, and customer demand vary across regions?
5. What KPIs should executives track on a daily, weekly, and monthly basis?
6. How can stakeholders interactively explore trends using filters, drill-throughs, and dynamic visuals?